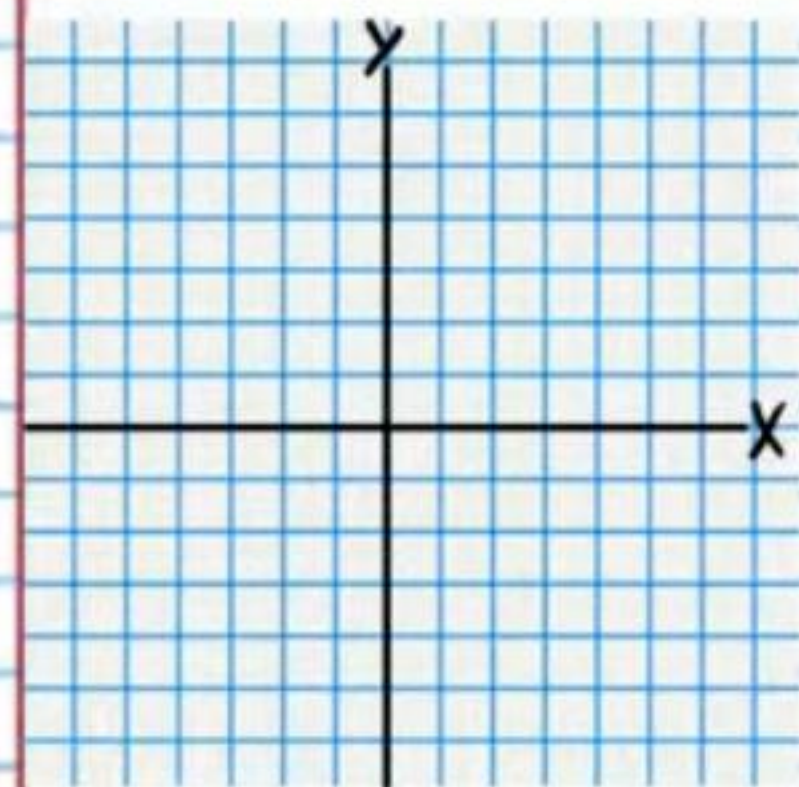


Geometry Final Test (Part II)

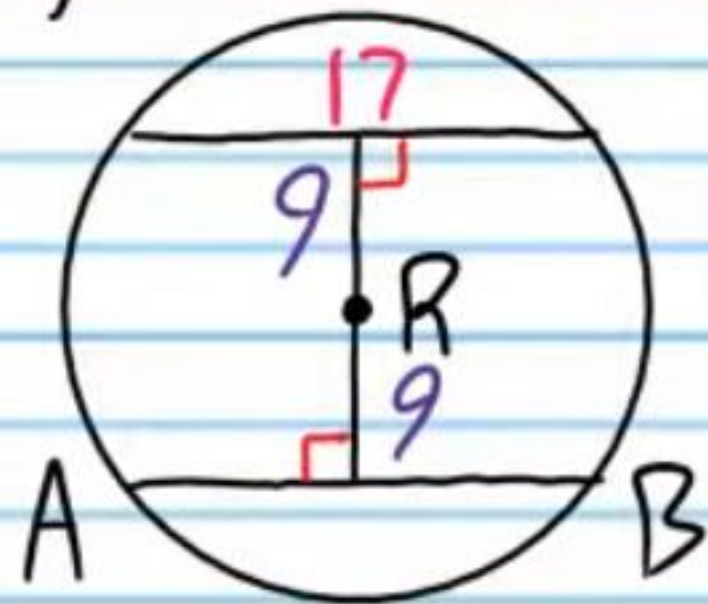
Name
Date
Course

- Write your name, the date, and the course title (i.e. Geometry).
- Show all work.
- You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
- If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- Graphing paper may be useful for this test.
- Each problem is worth four points for a total of 100 points.

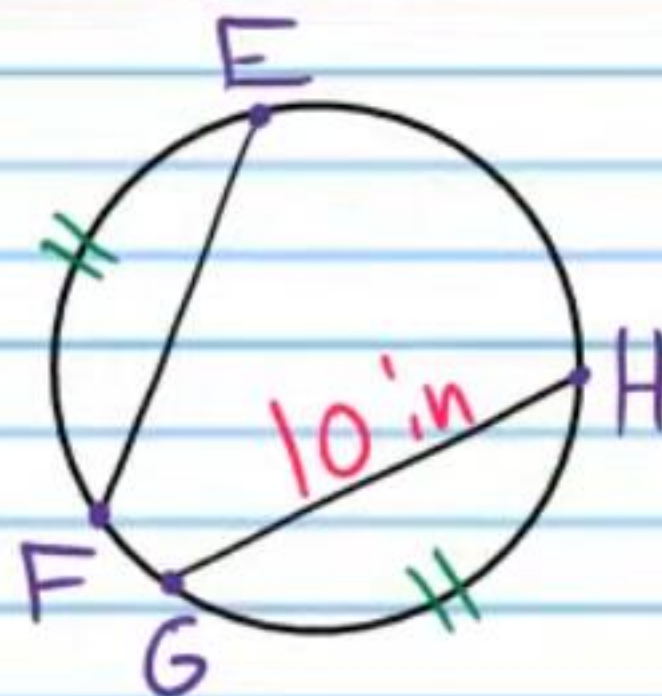
① Graph the following equation: $(x+2)^2 + (y-4)^2 = 1$



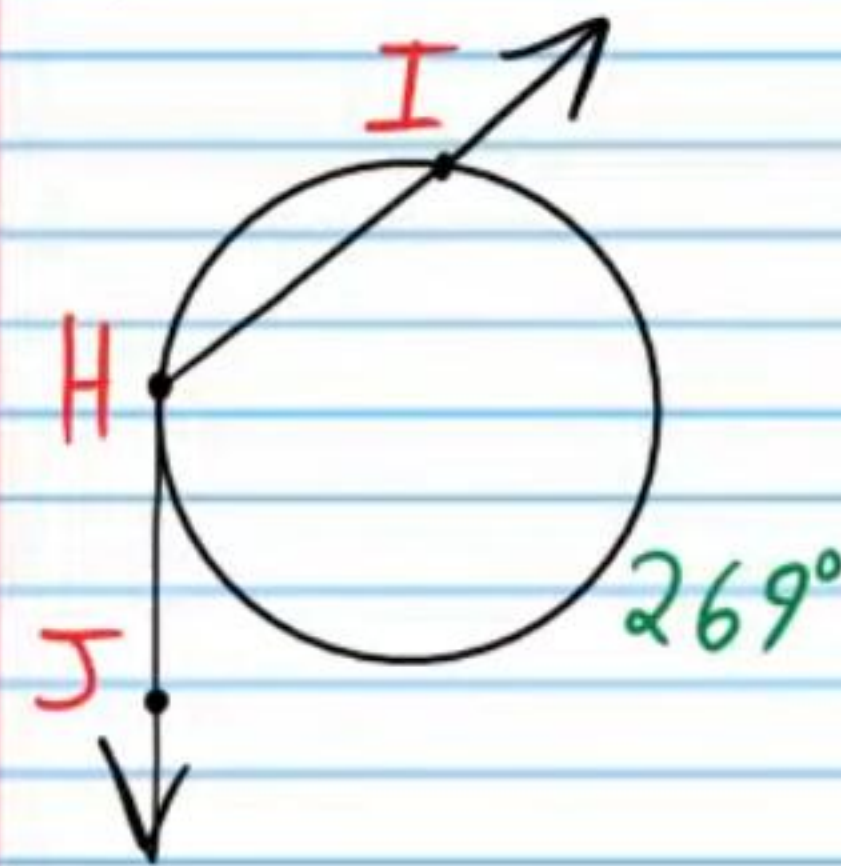
② i) Find AB in $\odot R$.



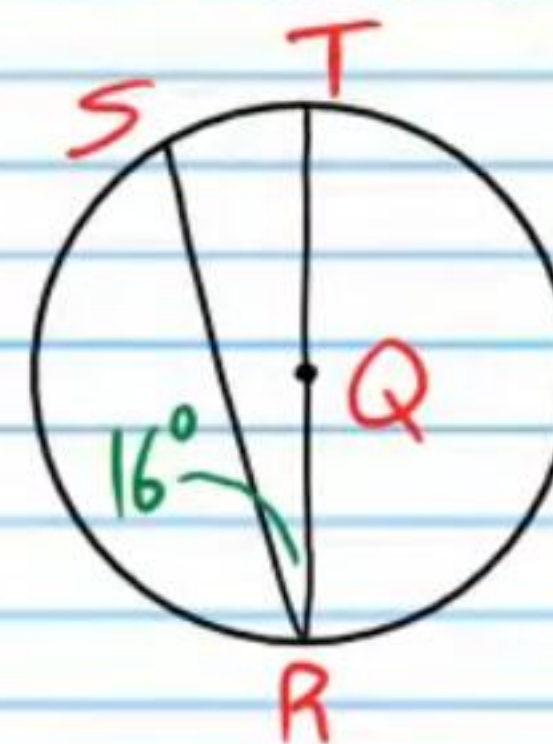
ii) Find EF .



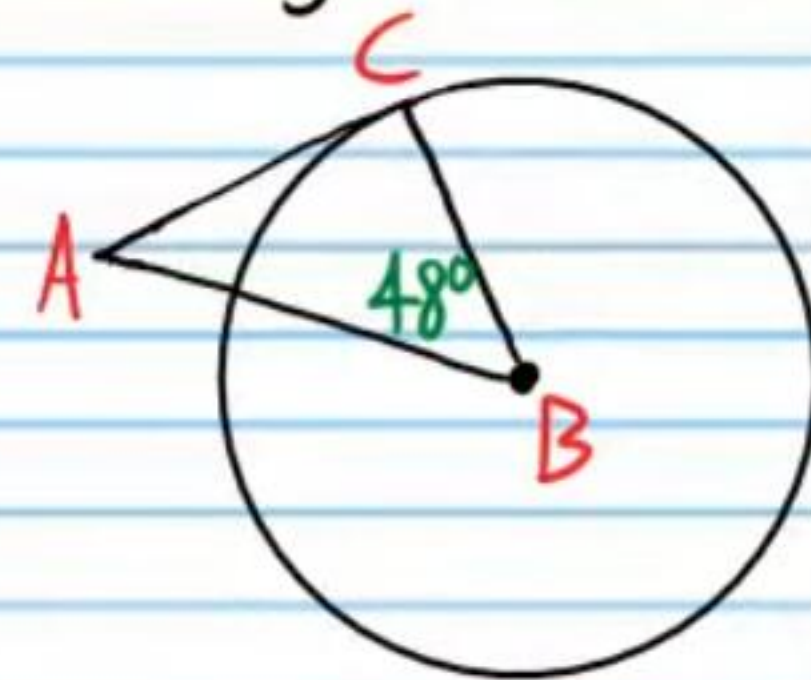
iii) Find $m\angle IHJ$.



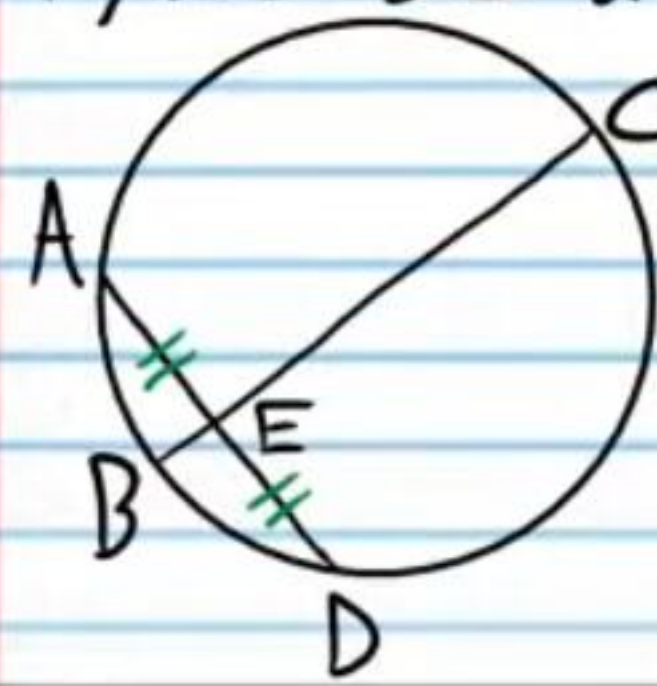
iv) Find $m\widehat{SRT}$ if Q is the center.



③ i) If the line containing $\overline{AC}$ is tangent to $\odot B$ at C, find $m\angle A$.

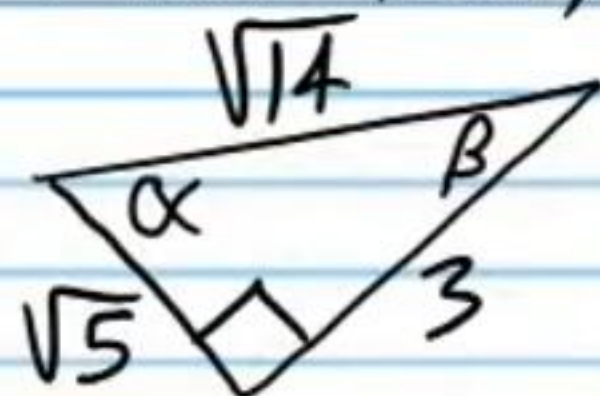


ii) Is $\overline{BC}$ a diameter? Explain.



④ i) Find the exact value of $\tan 60^\circ$ without a calculator. Hint: You must use a special triangle.

ii) Find $\tan \beta$



⑤ Find the unknown values in the parallelogram below.

i) a

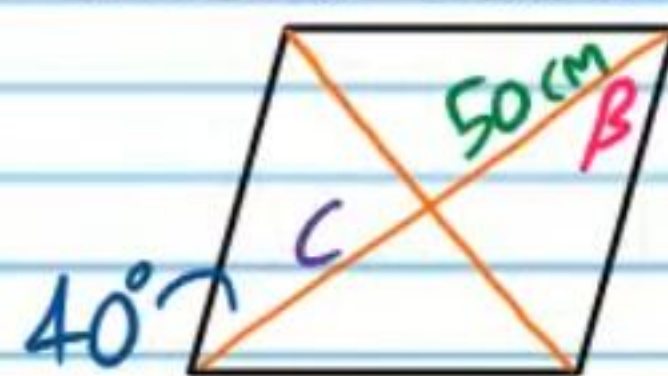
ii) b



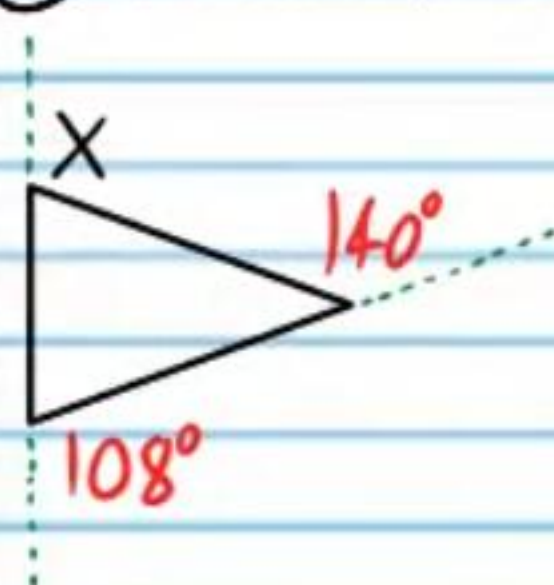
iii) c

iv) β

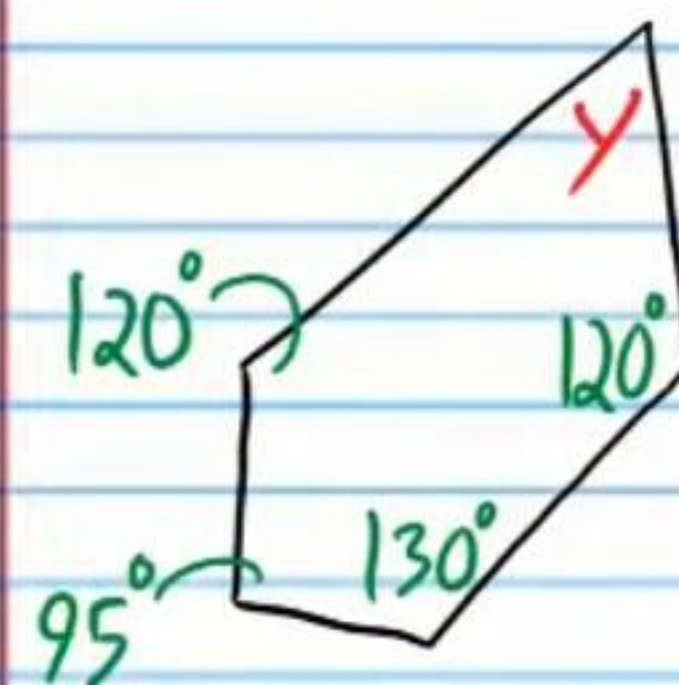
Find the unknown values in the rhombus below.



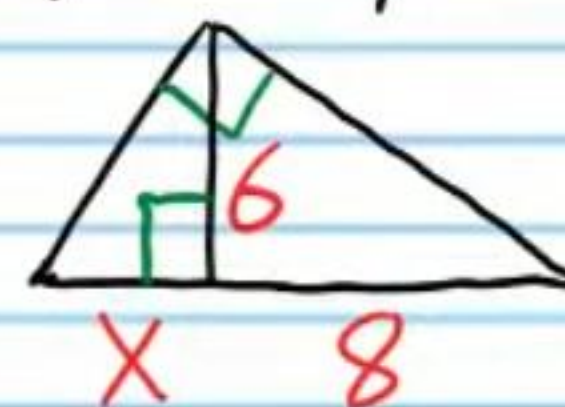
⑥ Find x .



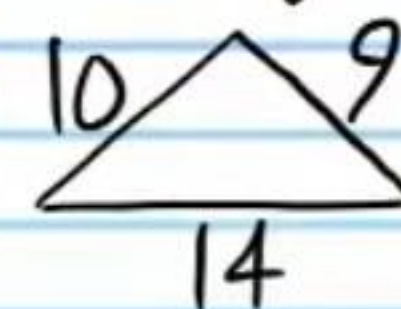
⑦ Find y .



⑧ i) Find y .



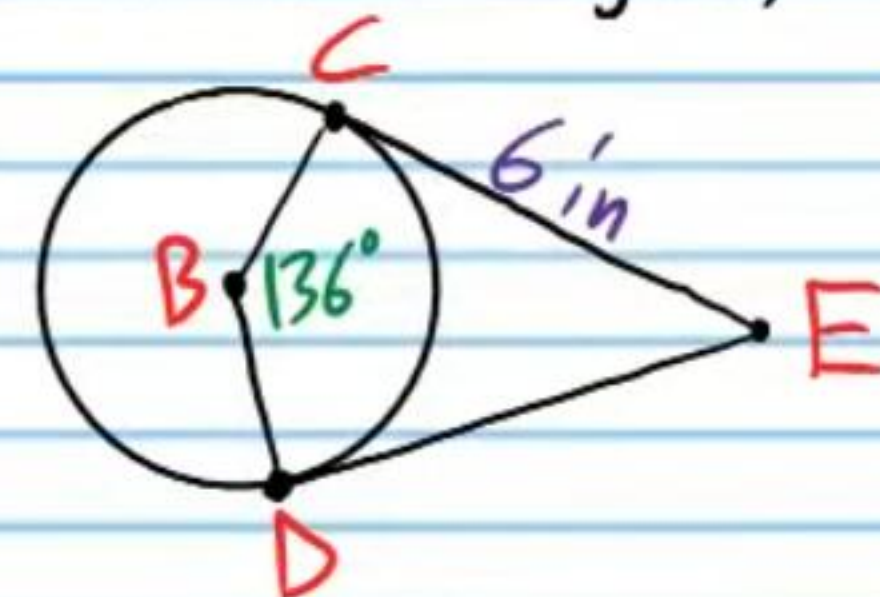
ii) Is the following polygon an acute, obtuse, or right triangle?



⑨ If the segments below are radii and tangents, find

i) DE

ii) $m\angle E$

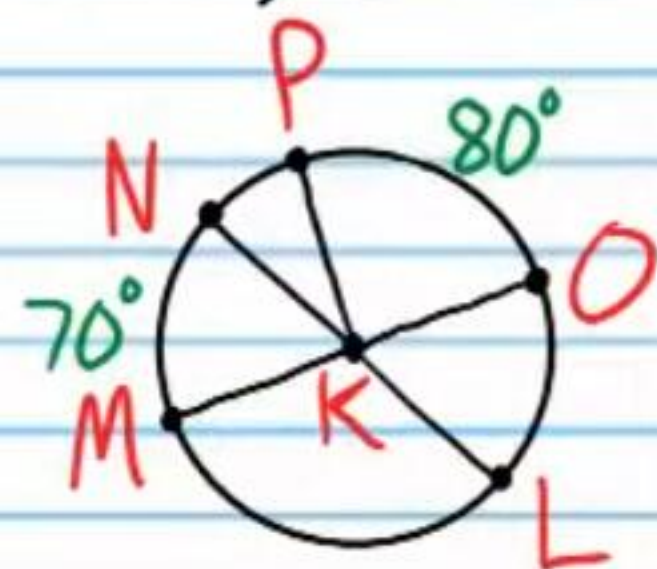


If $\overline{NL}$ and $\overline{MO}$ are diameters of $\odot K$ below, find

iii) $m\hat{NP}$

iv) $m\angle LKO$

Hint: Use the vert. $\angle$'s Thm.



10) Find the unknown values in the kite below.

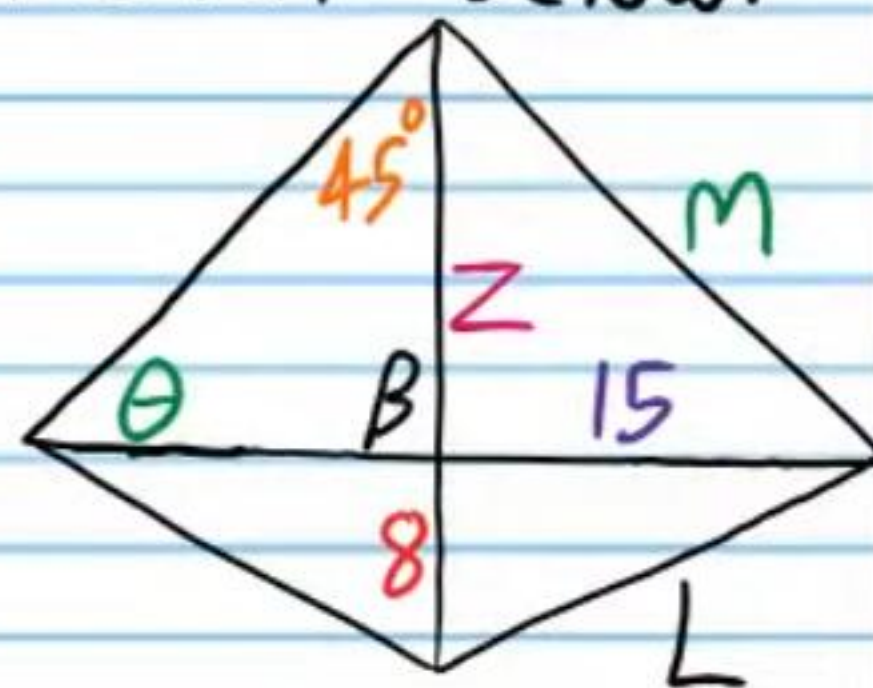
i) β

ii) θ

iii) z

iv) m

v) L



Find the unknown values in the trapezoid below.

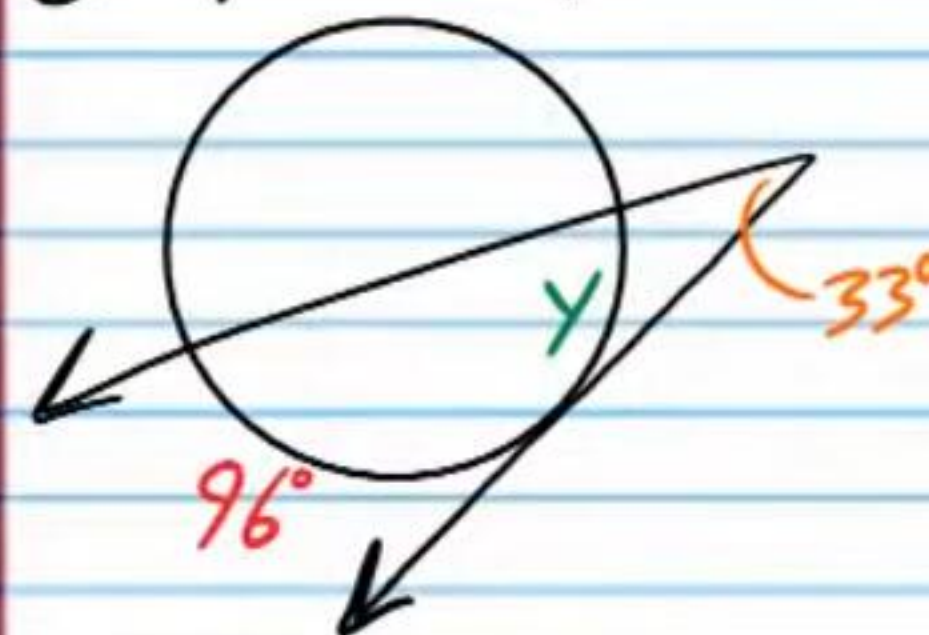
vi) α

vii) e

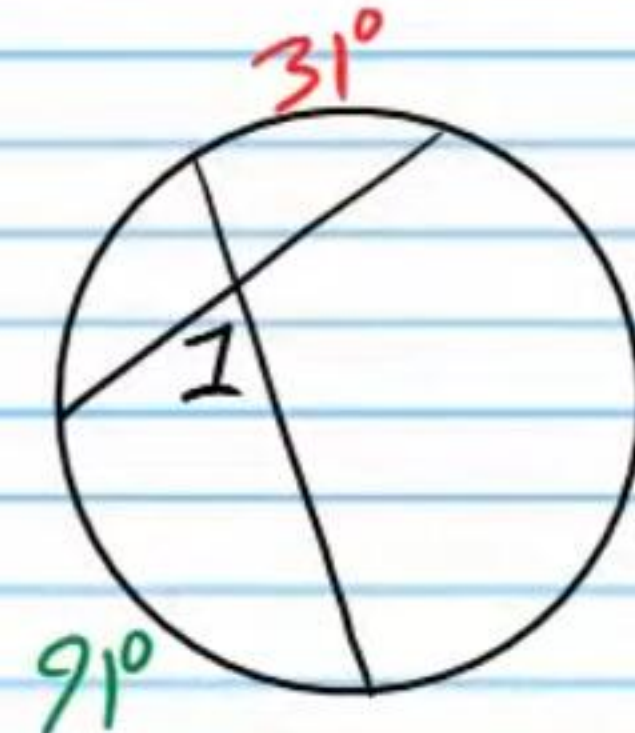
viii) p



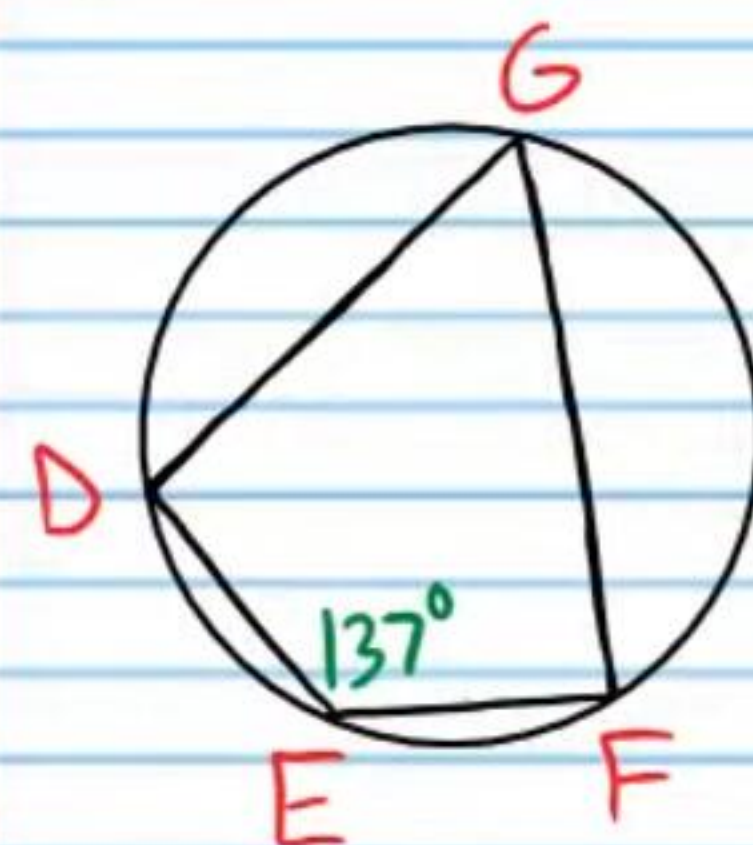
11) i) Find y .



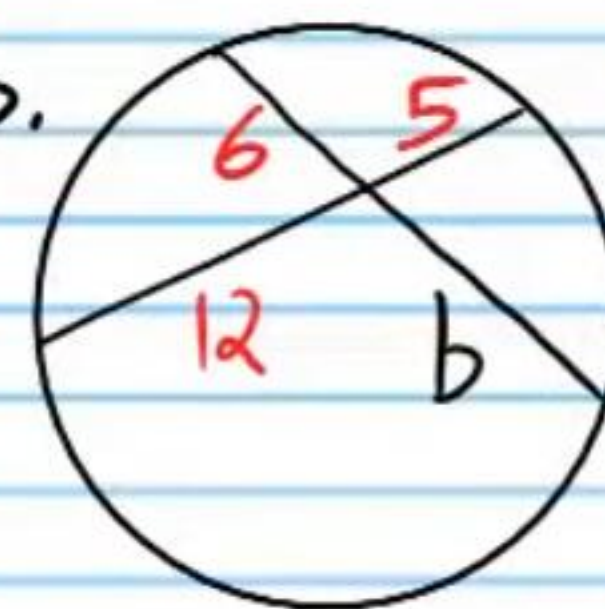
ii) Find $m\angle 1$.



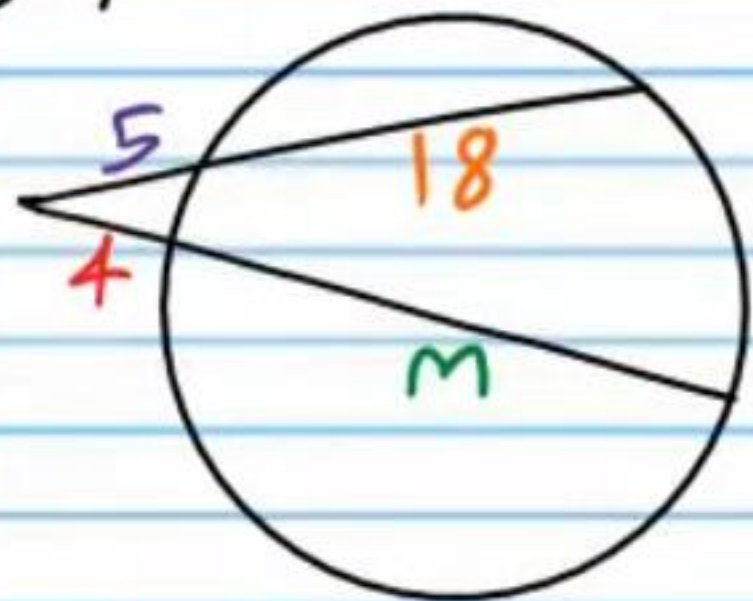
iii) Find $m\angle G$.



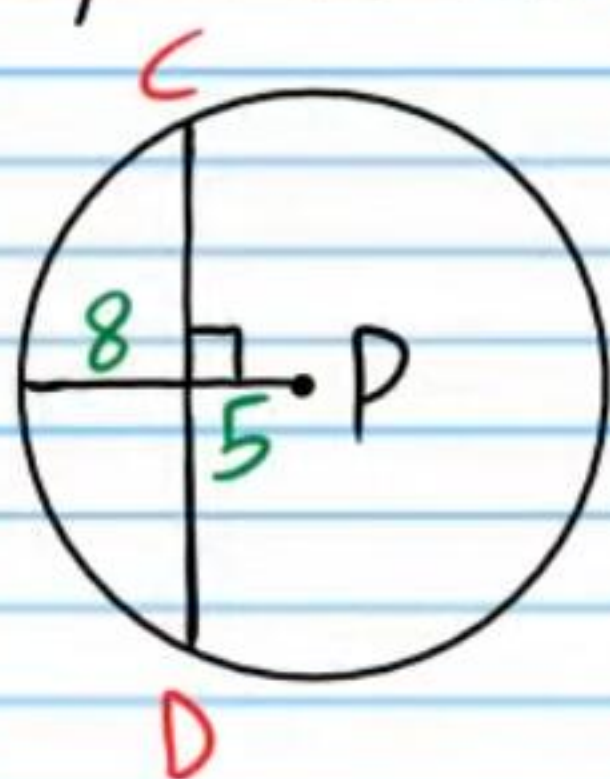
iv) Find b .



⑫ i) Find m .

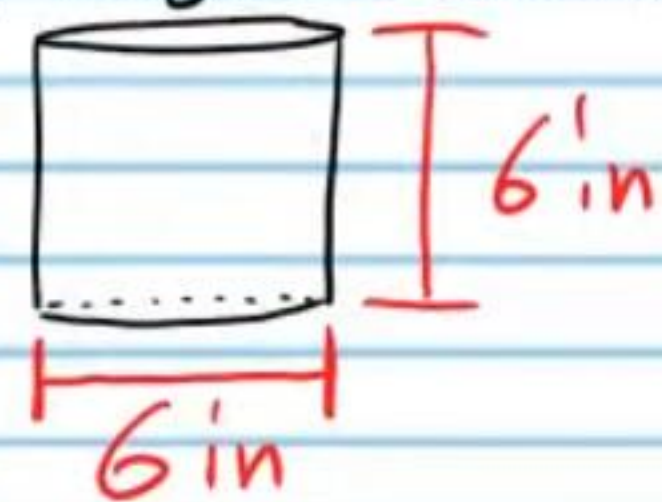


ii) Find CD in $\odot P$.

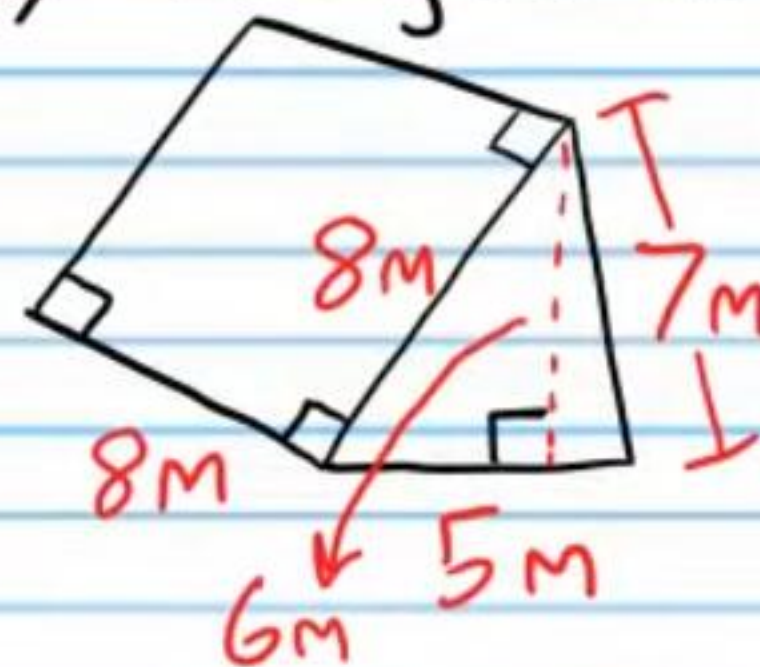


⑬ Find the surface area of the solids. Round to the nearest tenth.

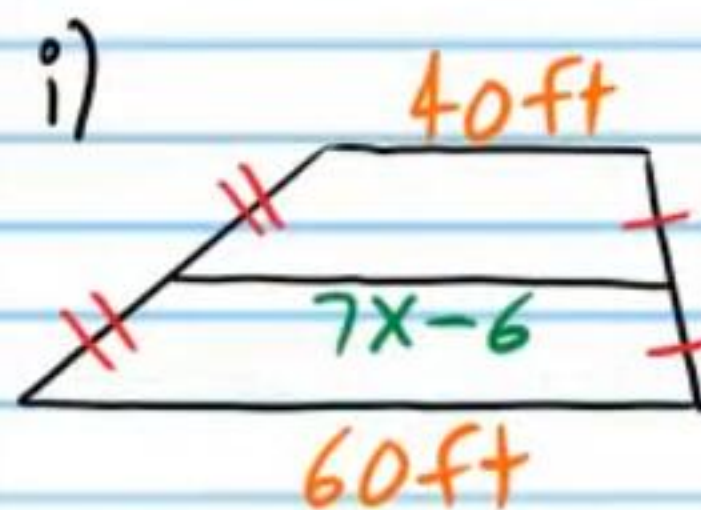
i) Right Circular Cylinder



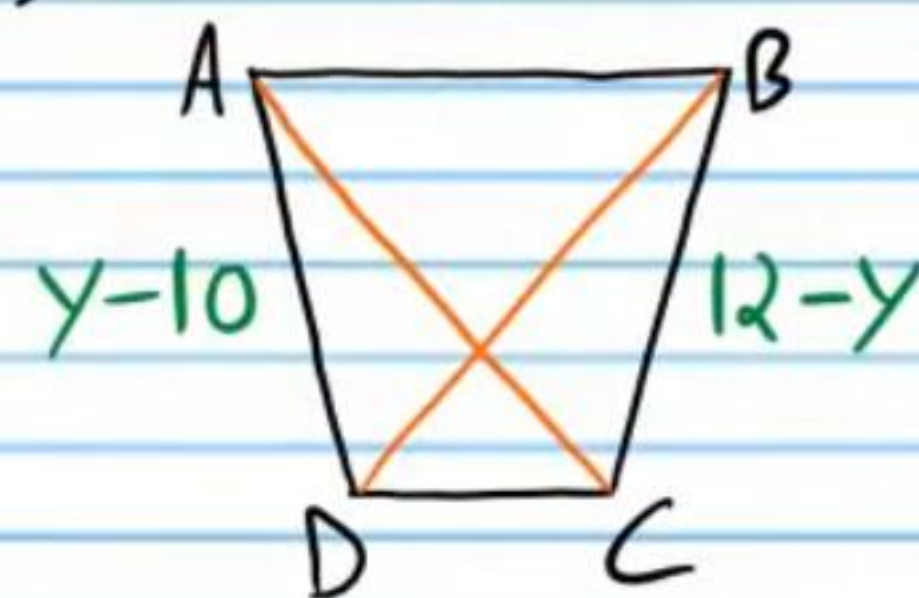
ii) Triangular Prism



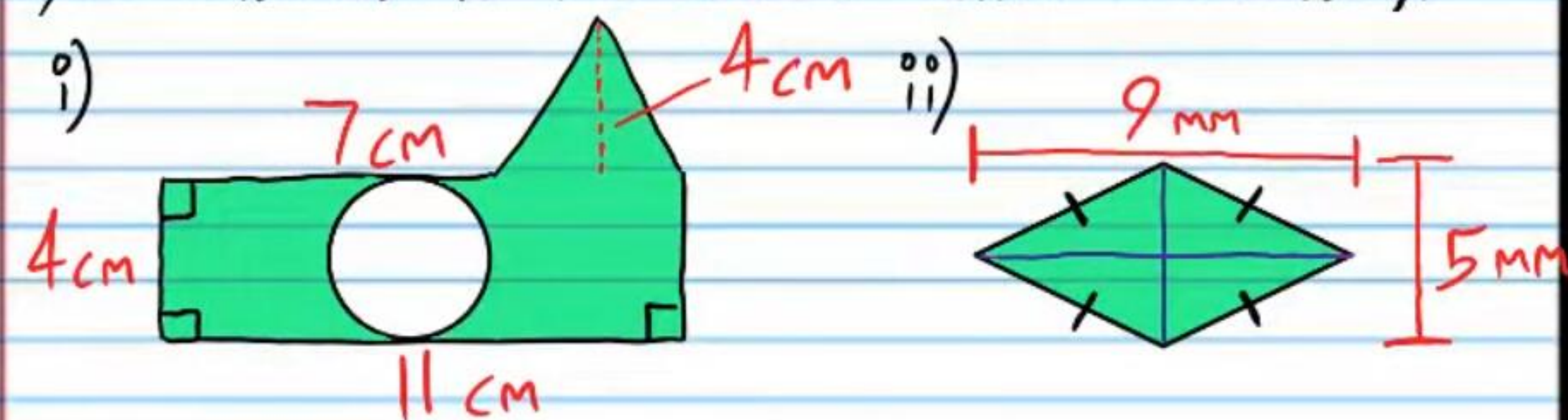
⑭ Find the unknown values in the trapezoids below.



ii) $\overline{AC} \cong \overline{BD}$



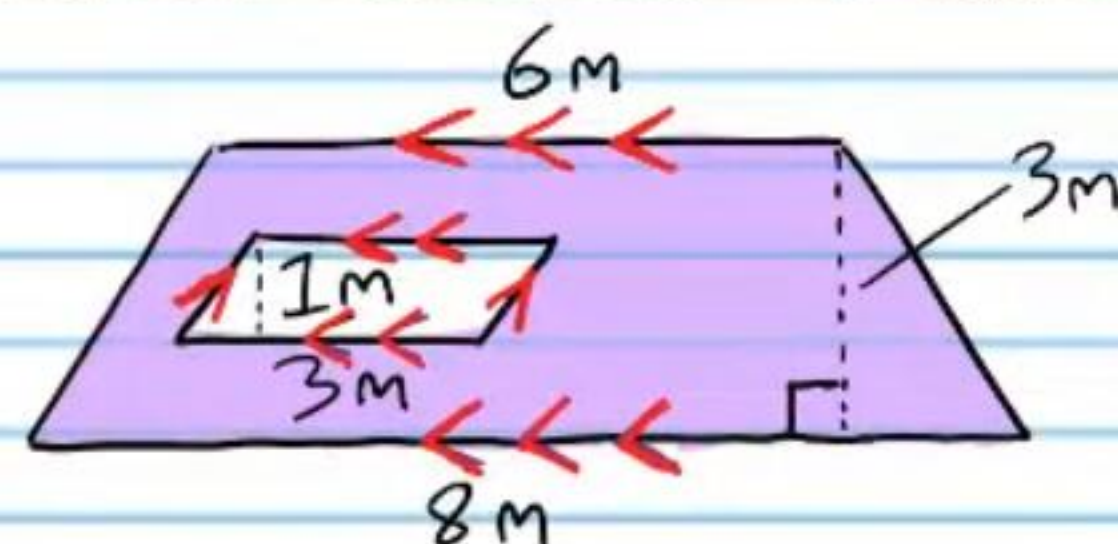
⑮ Find the area of each shaded region below. Round your answers to the nearest tenth if necessary.



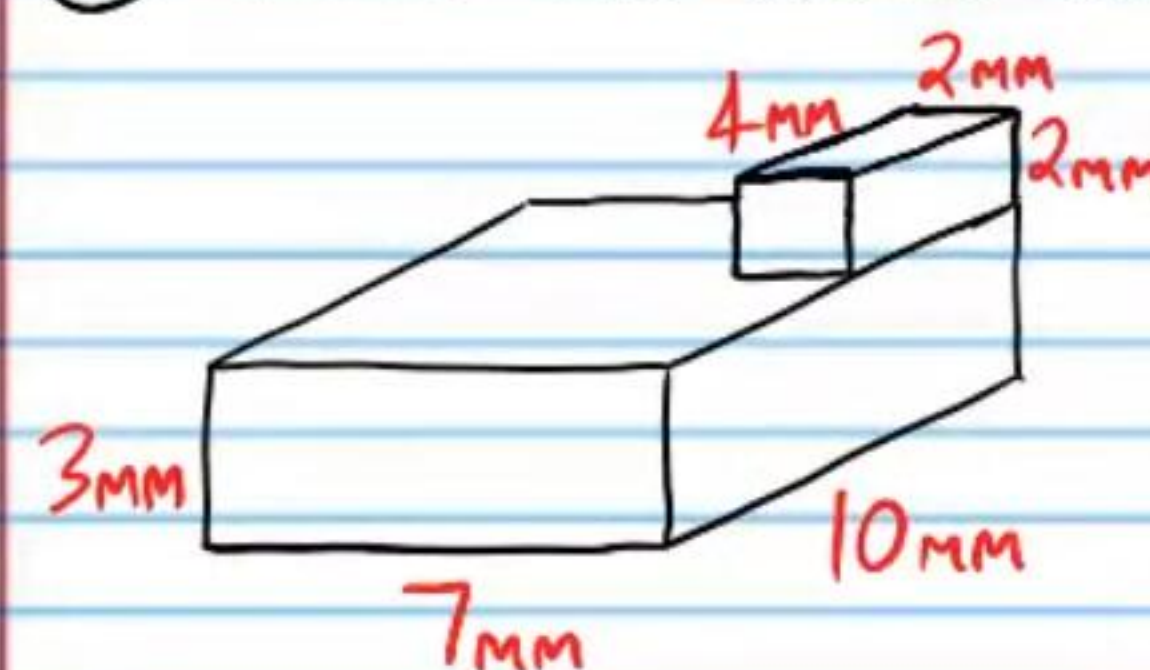
⑰ The ratio of corresponding sides (i.e. Scale factor) of two similar solids is written below. Given the Volume of the smaller solid, find the volume of the larger solid.

$$SF = 3, V = 2 \text{ m}^3$$

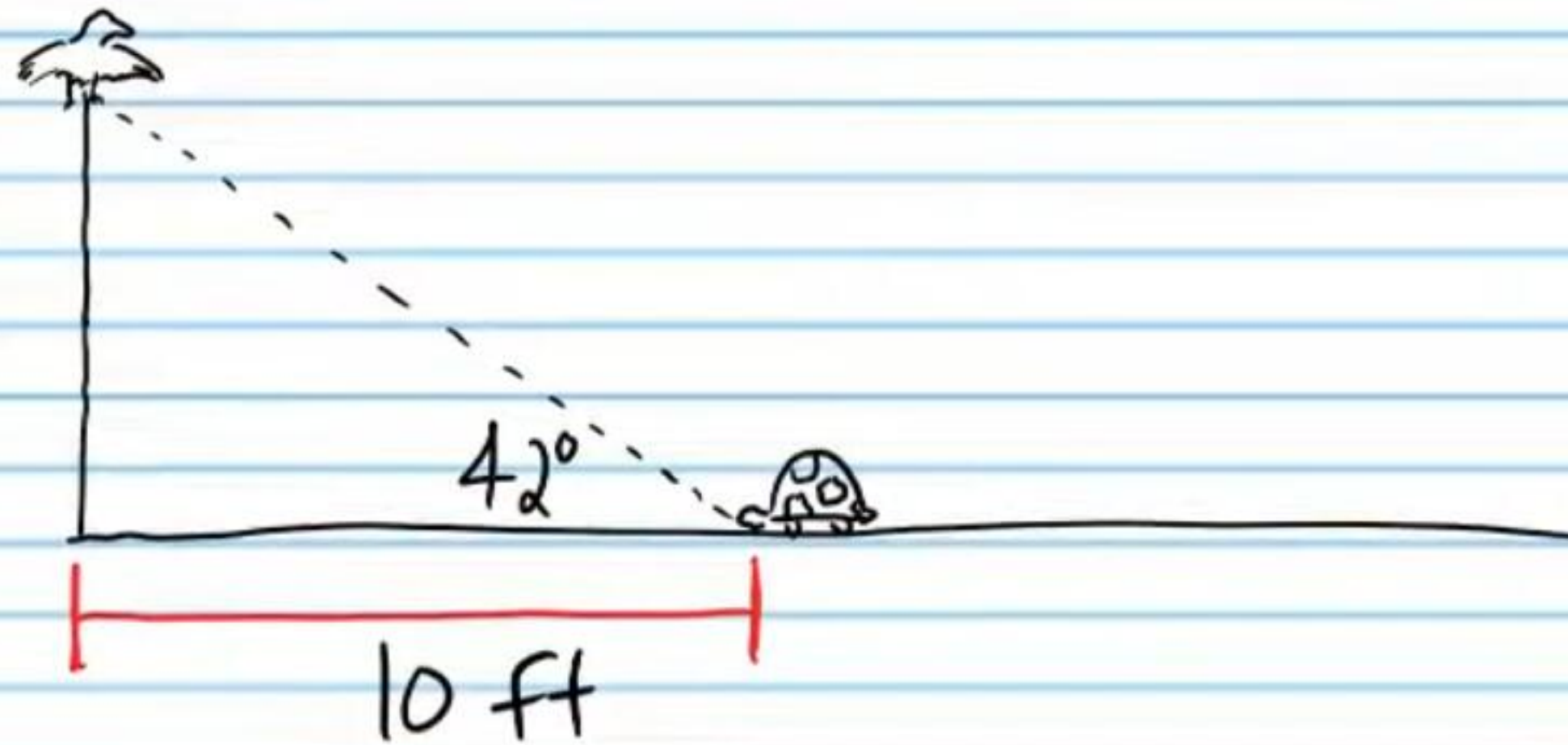
⑯ Find the probability that a randomly chosen point in the figure will lie in the shaded region. Write your answer as a fraction. Remember to reduce fractions.



⑱ Find the Volume of the following solid.

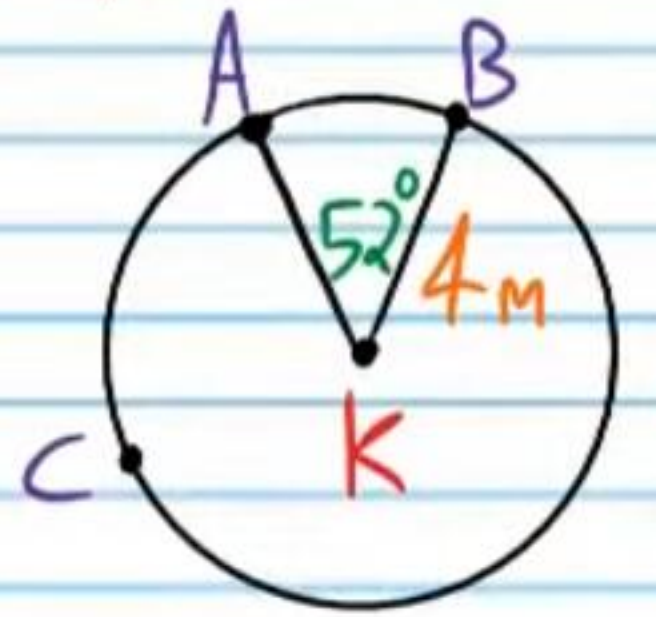


- ①9 A turtle looks up at a bird perched on a fence 10 feet away. If the turtle looks up, with an angle of elevation from the ground of 42° , how high is the bird from the ground? Round to the nearest hundredth.

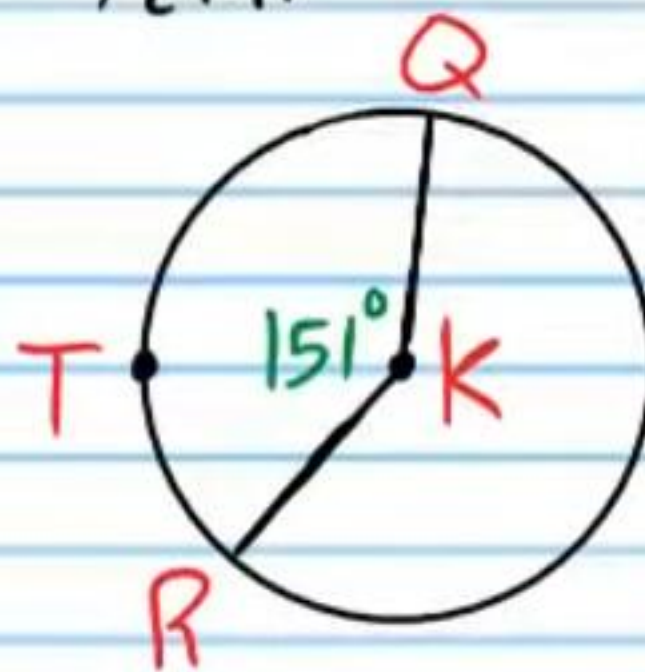


If K is the center of the circles below, find the following values. Round your answers to the nearest hundredth.

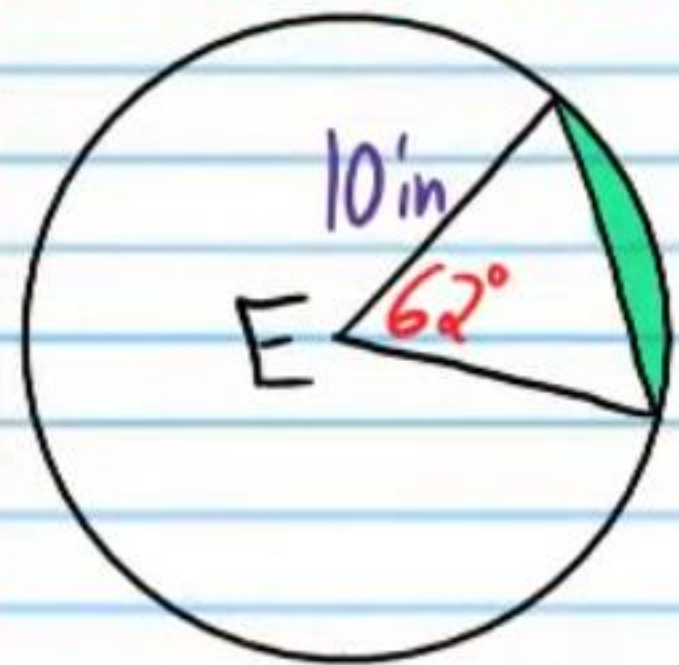
- ②0 $\angle ACB$



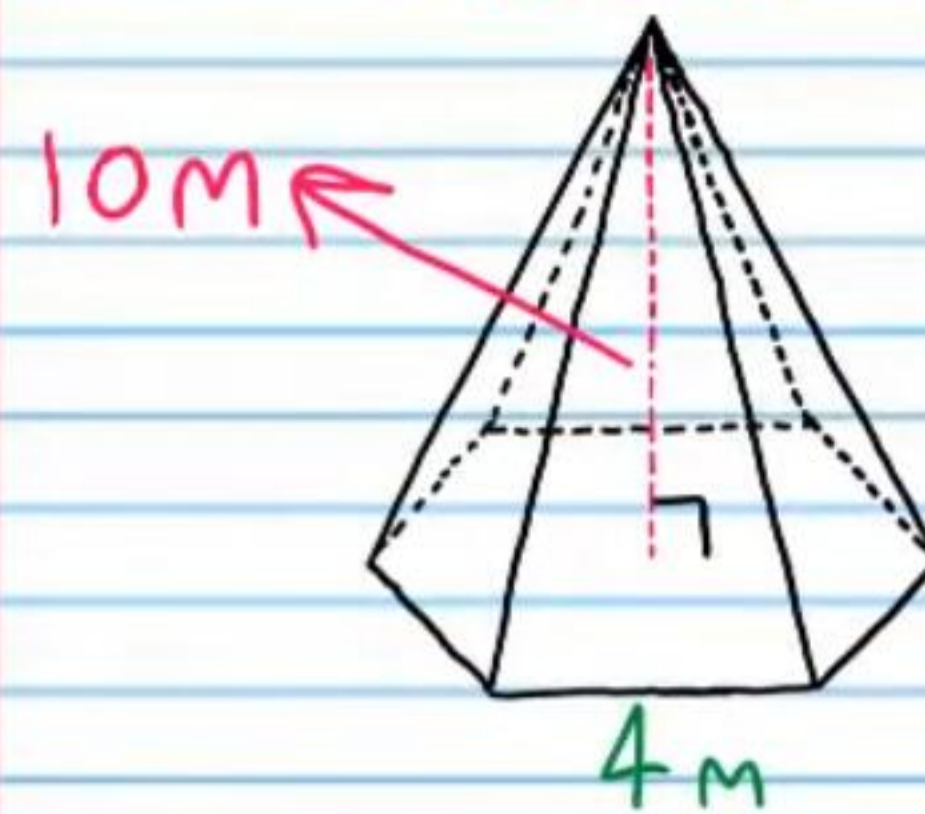
- ②1 The circumference if $\angle RTQ = 7\text{ cm}$.



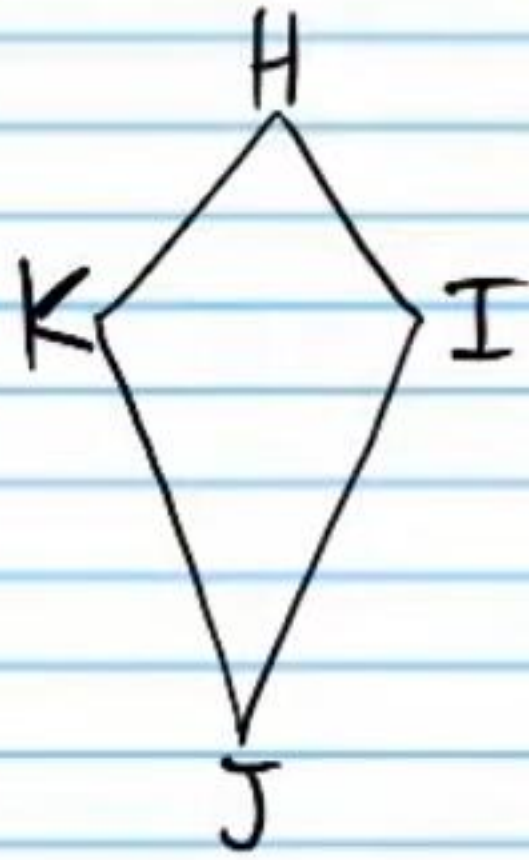
② Find the area of the shaded region in $\odot E$.
Round to the nearest hundredth.



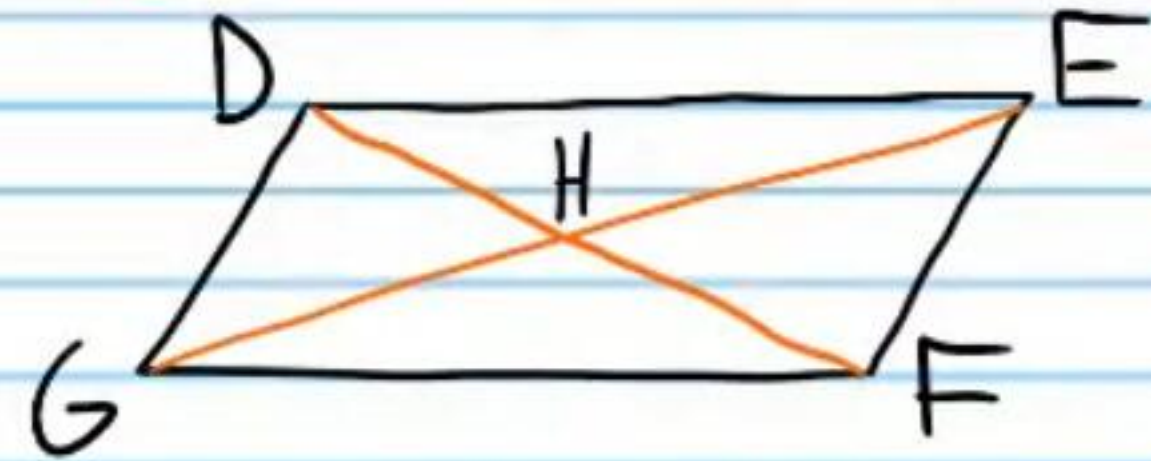
③ Find the volume of the regular pyramid below. Round to the nearest tenth.



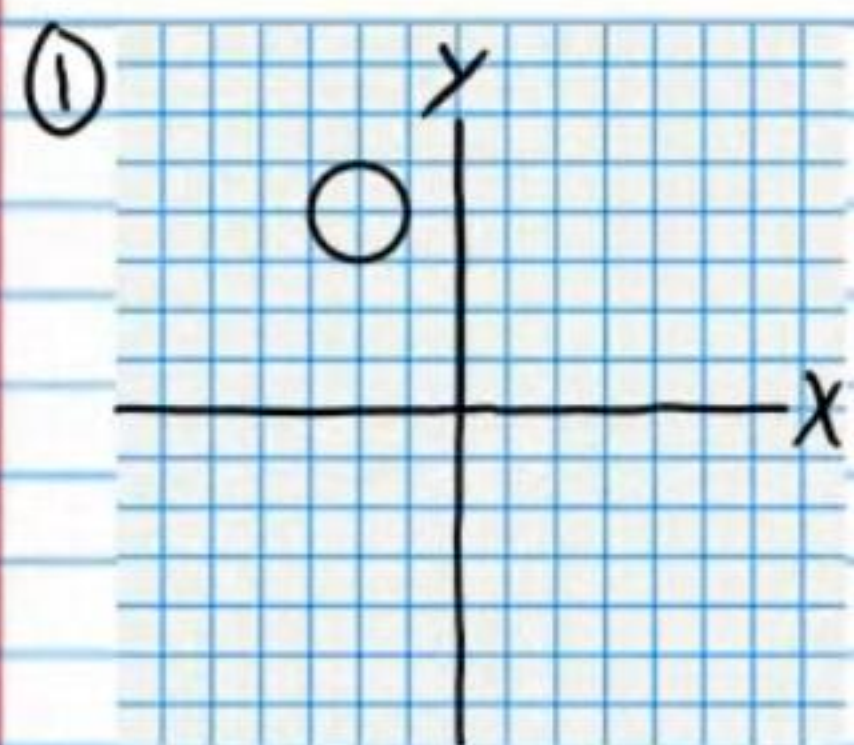
24 Write a paragraph proof showing that if $H I J K$ below is a kite, then $\angle K \cong \angle I$.



25 Write a flowchart proof showing that if the quadrilateral below is a parallelogram, then $\overline{DF}$ and $\overline{GE}$ bisect each other. You must use the Parallelogram Opposite Sides Theorem.



Geometry Final Test (Part II) Answers



② i) $AB = 17$

ii) $EF = 10$ in

iii) $m\angle IHJ = 134.5^\circ$

iv) $m\widehat{SRT} = 328^\circ$

③ i) $m\angle A = 42^\circ$

ii) Not enough information.
We do not know that $\overline{BC} \perp \overline{AD}$.

④ i) $\sqrt{3}$

ii) $\sqrt{5}/3$

⑤ i) $a = 137^\circ$

ii) $b = 43^\circ$

iii) $c = 50$ cm

iv) $\beta = 40^\circ$

⑥ $x = 112^\circ$

⑦ $y = 75^\circ$

⑧ i) $x = \frac{9}{2}$ or 4.5

ii) Obtuse

⑨ i) $DE = 6$ in

ii) $m\angle E = 44^\circ$

iii) $m\widehat{NP} = 30^\circ$

iv) $m\angle LKO = 70^\circ$

⑩ i) $\beta = 90^\circ$

ii) $\theta = 45^\circ$

iii) $z = 15$

iv) $m = 15\sqrt{2}$

v) $L = 17$

vi) $\alpha = 132^\circ$

vii) $e = 48^\circ$

viii) $P = 132^\circ$

⑪ i) $y = 30^\circ$

ii) $m\angle I = 61^\circ$

iii) $m\angle G = 43^\circ$

iv) $b = 10$

⑫ i) $m = 24.75$

ii) $CD = 24$

⑬ i) 169.6 in²

ii) 190 m²

⑭ i) $x = 8$

ii) $y = 11$

⑮ i) 39.4 cm²

ii) 22.5 mm²

⑯ $\frac{6}{7}$

⑰ 54 m³

⑱ 226 mm³

⑲ 9 ft

⑳ $\angle ACB = 21.5$ m

㉑ 16.69 cm

(22) 9.96 in^2

(23) 138.6 m^3

★ Slight Variations of this proof are okay.

(24) It is given that $H I J K$ is a kite. By the definition of a kite, $\overline{KH} \cong \overline{HI}$ and $\overline{KJ} \cong \overline{JI}$. We also know that the auxiliary line segment $\overline{HJ} \cong \overline{HJ}$ by the Reflexive Property of Congruence. Consequently, $\triangle KHJ \cong \triangle I H J$ by the SSS Postulate and because corresponding parts of congruent triangles are congruent, we can conclude that $\angle K \cong \angle I$.

(25) $DEFG$ is a parallelogram

★ Slight Variations of this proof are okay.

Given

$\overline{DG} \parallel \overline{EF}$ & $\overline{DE} \parallel \overline{GF}$

Def. of 

$\angle GDF \cong \angle EFD$
& $\angle GEF \cong \angle DGE$

AIAT

$\overline{GD} \cong \overline{EF}$

POST

$\triangle GDH \cong \triangle EFH$

ASA

$\overline{DH} \cong \overline{HF}$ & $\overline{GH} \cong \overline{HE}$

CPCTC

$\overline{DF}$ & $\overline{GE}$ bisect each other.

Def. of Seg Bisector.